

Android Studio: A Platform for Android Development

Android Studio is an integrated development environment (IDE) for the Android platform. It simplifies app development. Though offered by Google, seasoned Java developers will immediately recognise that the toolkit is a version of IntelliJ IDEA.

According to IDC, globally, Android's share of the smartphone market is about 45 per cent. The best part is that Android is open source and learning it is not at all difficult. Students and professionals want to, at least, know its basics. There are many platforms, like Android Studio, where even beginners can get into Android development. Android Studio is a cross-platform integrated development environment (IDE) for developing on the Android platform. It is written in Java and is available for Linux, Windows as well as for Mac OS X.

Eclipse, which also provided Android development tools, has been replaced by Android Studio as Google's primary IDE for native Android application development. The main reason for this move is because Eclipse was not stable.

Android Studio offers a better *Gradle* build environment, smarter short cuts, an improved user interface (UI) designer, a better memory monitor, an improved string translation editor and better speed. The build system in Android Studio replaces the Ant system used with Eclipse ADT. It can run from the menu as well as from the command line. It allows you to track memory allocation as it monitors memory use. It has built-in support for the Google Cloud Platform, making it easy to integrate Google Cloud Messaging and App Engine. It also comes with inline debugging, and performance analysis tools.

Android Studio has Android Virtual Device (AVD) which comes with emulators for Nexus 6 and Nexus 9 devices. It also offers build variants and the capability to generate multiple *apk* files. Whenever one compiles a program, the configured *lint* and IDE inspections run automatically.

Configuration

Installation

Before you set up Android Studio in Linux, you need to install JDK 6 or higher. In fact, JDK 7 is required for developing Android 5.0 or above. The other requirements are a minimum of 2GB RAM (though 4GB is recommended), 400MB hard disk space and at least 1GB for the Android SDK, emulator system images, caches, GNU C Library (glibc) 2.15 or later, etc. After installing Android Studio and setting it up, go to the SDK manager to update the required tools, platforms, etc, required for app-building. These packages provide the basic SDK tools for app development, without an IDE. If you prefer to use a different IDE, the standalone Android SDK tools can be downloaded. One can set up an update channel to *Stable* by going to: *File > Settings > Appearance & Behaviour System Settings > Updates* as shown in Figure 1.